Introduction

Amir H. Payberah
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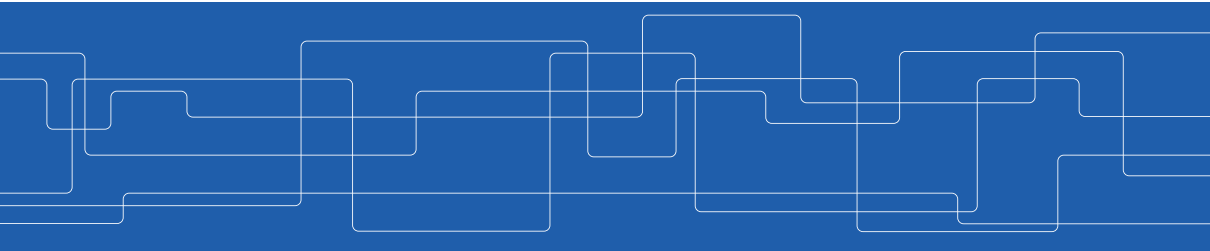
2026-10-28



Course Information

Course Objective

- This course has a system-based focus.

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- Learn the theory of machine learning and deep learning.

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- Learn the theory of machine learning and deep learning.
- Learn the practical aspects of building machine learning and deep learning algorithms using platforms, such as TensorFlow and PyTorch.

Intended Learning Outcomes (ILOs)

- ILO1: explain the principles of ML/DL algorithms and apply their techniques to solve problems.

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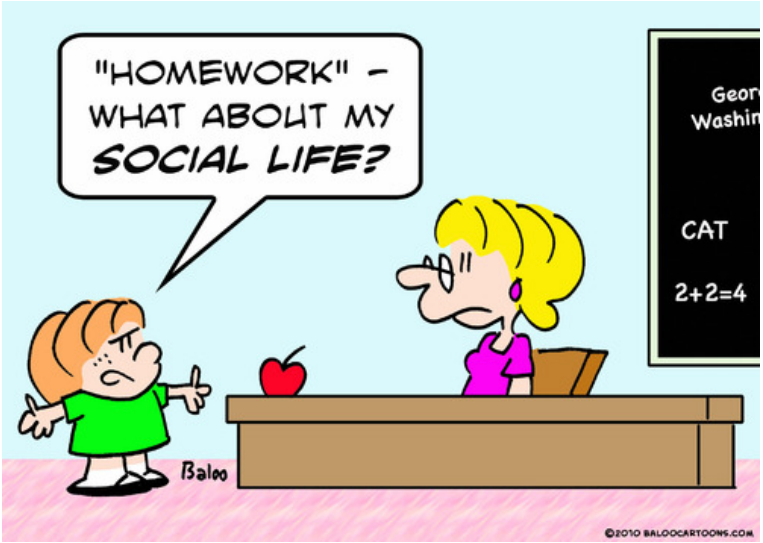
- ILO1: explain the principles of ML/DL algorithms and apply their techniques to solve problems.
- ILO2: demonstrate an ability to design and implement scalable ML/DL systems.
- ILO3: critically analysis ML/DL systems.


"HOMEWORK" – WHAT ABOUT MY SOCIAL LIFE?
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2+2=4
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The Course Assessment

- Task1: two lab assignments (P/F).

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- Task2: the essay and the presentation (P/F).
- Task3: the project assignment (P/F).
- Task4: the final exam (A-F).
- Task1, Task2 and Task3 should be done in groups of three students.

How Each ILO is Assessed?

	Task1	Task2	Task3	Task4
ILO1	x		x	x
ILO2	x		x	
ILO3		x		

Task1: The Lab Assignments (P/F)

- Two lab assignments: CNN and LLM
- What to deliver: the source code and report.

Task2: The Essay and The Presentation (P/F)

- Select one paper at the intersection of scalable ML/DL systems and societal/environmental issues.
- What to deliver: an essay and recorded video (10-15 minutes).

Task3: The Final Projectt (P/F)

- Proposed by students and confirmed by the teacher.
- What to deliver: the source code and report.

Task4: The Final Exam (A-F)

- Questions about the lectures and the labs.

The Final Grade

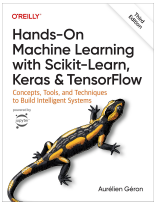
- LAB1 - Programming Assignments, 3.0 credits (P/F)
  - You must successfully complete Task1, Task2, and Taks3.
- TEN1 - Examination, 4.5 credits (A-F)

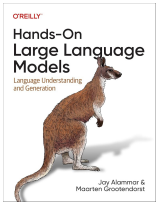
How to Submit the Assignments?

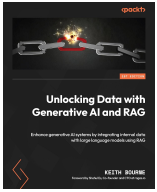
- Through the Canvas site.
- Students will work in groups of three on Task1, Task2, and Task3.

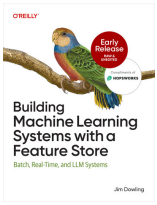
The Course Material

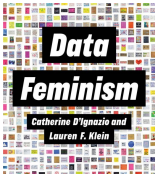
- Hands-on Machine Learning with Scikit-Learn and TensorFlow, A. Geron, O'Reilly Media, 2022.
- Hands-on Large Language Models: Language Understanding and Generation, J. Alammar and M. Grootendorst, O'Reilly Media, 2024.
- Unlocking Data with Generative AI and RAG: Enhance Generative AI Systems by Integrating Internal Data with Large Language Models using RAG, K. Bourne, Packt Publishing Ltd, 2024.
- Building ML Systems with a Feature Store), J. Dowling, O'Reilly Media, 2025.
- Data Feminism, C. D'ignazio and L. Klein, MIT press, 2020.











The Course Overview

Artificial Intelligence Challenge

- Artificial intelligence (AI) can solve problems that can be described by a list of formal mathematical rules.

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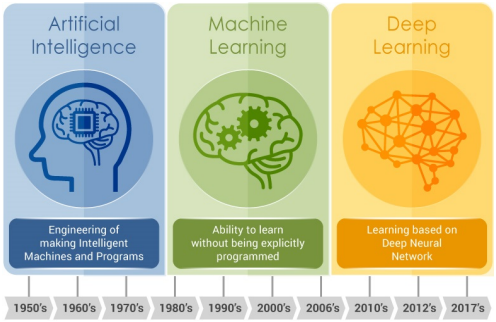
- The challenge is to solve the tasks that are hard for people to describe formally.
- Let computers to learn from experience.

AI Generations

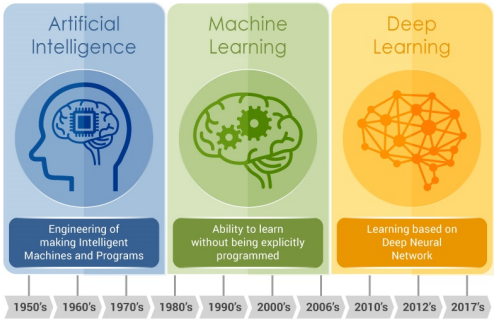
- Rule-based AI
- Machine learning
- Deep learning



[https://bit.ly/2woLEzs]

AI Generations - Rule-based AI

- Hard-code knowledge
- Computers reason using logical inference rules



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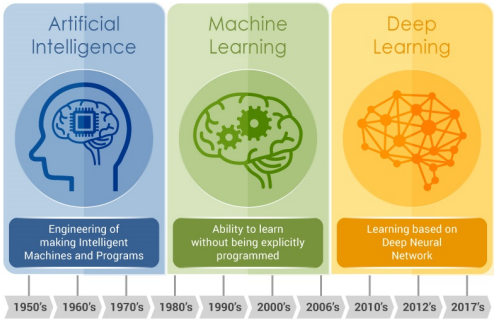
AI Generations - Machine Learning

- If AI systems acquire their own knowledge
- Learn from data without being explicitly programmed



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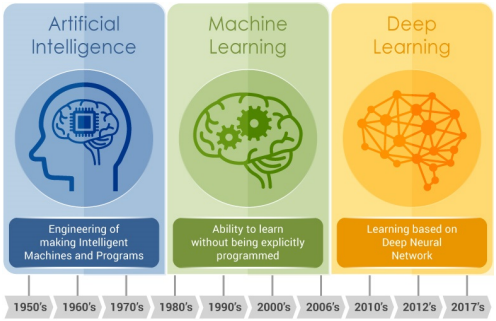
AI Generations - Deep Learning

- For many tasks, it is difficult to know what features should be extracted
- Use machine learning to discover the mapping from representation to output



[https://bit.ly/2woLEzs]

Sheepdog or Mop

Chihuahua or Muffin

Barn Owl or Apple

Why Does Deep Learning Work Now?

- Huge quantity of data
- Tremendous increase in computing power
- Better training algorithms

Data

GPUs

Weight Initialization

Non-Linearity

Machine Learning and Deep Learning

Learning Algorithms

- A ML algorithm is an algorithm that is able to learn from data.
- What is learning?

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- A ML algorithm is an algorithm that is able to learn from data.
- What is learning?
- A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E. (Tom M. Mitchell)

Learning Algorithms - Example 1

- A spam filter that can learn to flag spam given examples of spam emails and examples of regular emails.

[https://bit.ly/2oiplYM]

Learning Algorithms - Example 1

- A spam filter that can learn to flag spam given examples of spam emails and examples of regular emails.
- Task T: flag spam for new emails
- Experience E: the training data
- Performance measure P: the ratio of correctly classified emails

[https://bit.ly/2oiplYM]

Learning Algorithms - Example 2

- Given dataset of prices of 500 houses, how can we learn to predict the prices of other houses, as a function of the size of their living areas?

[https://bit.ly/2MyiJUy]

Learning Algorithms - Example 2

- Given dataset of prices of 500 houses, how can we learn to predict the prices of other houses, as a function of the size of their living areas?
- Task T: predict the price
- Experience E: the dataset of living areas and prices
- Performance measure P: the difference between the predicted price and the real price

[https://bit.ly/2MyiJUy]

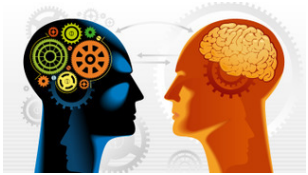
Types of Machine Learning Algorithms

- Supervised learning

- Unsupervised learning

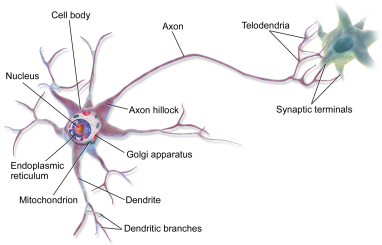
Types of Machine Learning Algorithms

- Supervised learning
  - Input data is labeled, e.g., spam/not-spam or a stock price at a time.
  - Regression vs. classification
- Unsupervised learning
  - Input data is unlabeled.
  - Find hidden structures in data.



From Machine Learning to Deep Learning

- Deep Learning (DL) is part of ML methods based on learning data representations.
- Mimic the neural networks of our brain.



[A. Geron, O'Reilly Media, 2017]

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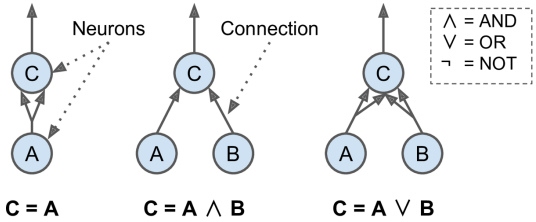
- Activates its output when more than a certain number of its inputs are active.



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The Linear Threshold Unit (LTU)

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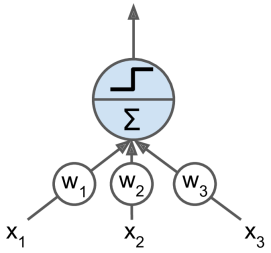
- Computes a weighted sum of its inputs and applies a step function to that sum.



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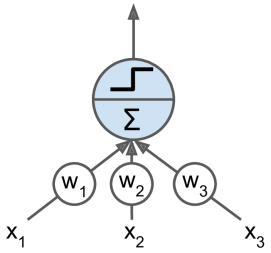
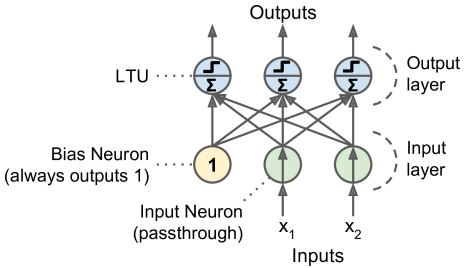
- $z = w_1 x_1 + w_2 x_2 + \cdots + w_n x_n = \mathbf{w}^\intercal \mathbf{x}$
- $\hat{y} = \text{step}(z) = \text{step}(\mathbf{w}^\intercal \mathbf{x})$



The Perceptron

- The perceptron is a single layer of LTUs.
- The input neurons output whatever input they are fed.
- A bias neuron, which just outputs 1 all the time.

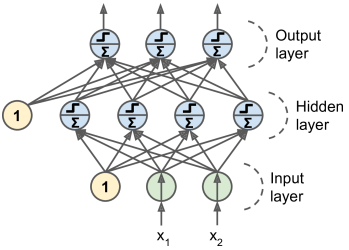


Deep Learning Models

- Deep Neural Network (DNN)
- Convolutional Neural Network (CNN)
- Recurrent Neural Network (RNN)
- Transformers and Large Language Models (LLM)

Deep Neural Networks

- Multi-Layer Perceptron (MLP)
  - One input layer.
  - One or more layers of LTUs (hidden layers).
  - One final layer of LTUs (output layer).



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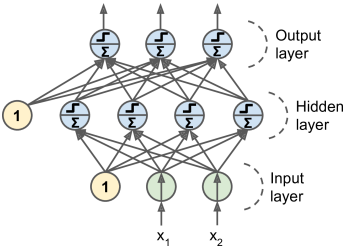

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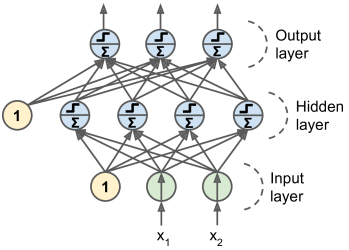
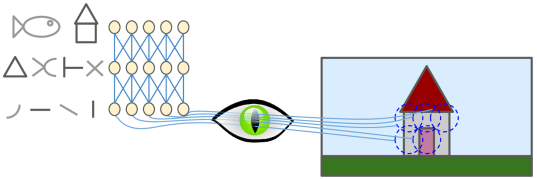
- Backpropagation training algorithm.



Convolutional Neural Networks

- Many neurons in the visual cortex react only to a limited region of the visual field.



Convolutional Neural Networks

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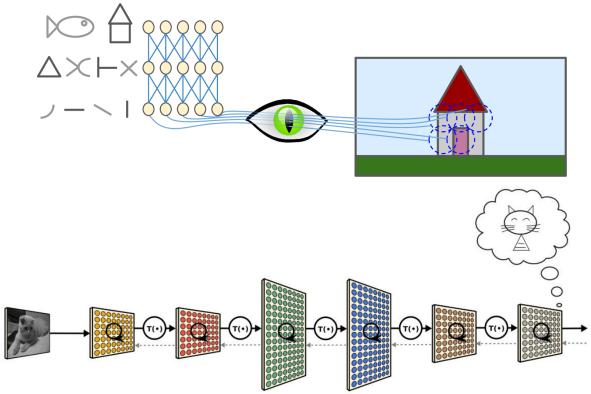
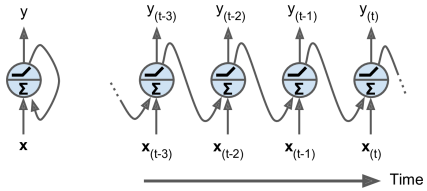
- The higher-level neurons are based on the outputs of neighboring lower-level neurons.



Recurrent Neural Networks

- The output depends on the input and the previous computations.



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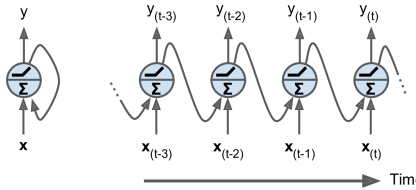

- Analyze time series data, e.g., stock market, and autonomous driving systems.
- Work on sequences of arbitrary lengths, rather than on fixed-sized inputs.

- Understanding LLMs

- Understanding LLMs
- Using LLMs

Transformers and Large Language Models

- Understanding LLMs
- Using LLMs
- Training and Fine-tuning LLMs

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- Retrieval-Augmented Generation (RAG)

MLOps and LLMOps

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- LLMOps: Applying these practices to LLM applications.
- Both tack experiments, test quality, manage versions, and monitor performance.
- LLMOps also manage prompts, retrieved documents, answer reliability, and token costs.

The Dual Lens of This Course

- Technical lens
- Critical lens

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- Technical lens
  - Understand ML/DL systems
  - Learn to design and deploy
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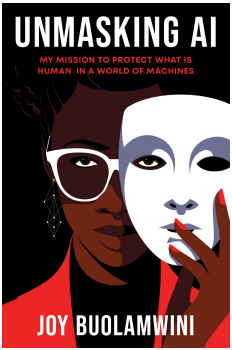
- Technical lens
  - Understand ML/DL systems
  - Learn to design and deploy
- Critical lens
  - Who controls and benefits from this infrastructure?
  - Reveal hidden labor
  - Track environmental costs: energy, water, e-waste

Why the Critical Lens Matters

- Technology is never neutral
  - E.g., facial recognition often misidentifies women and people with darker skin.



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- Technology is never neutral
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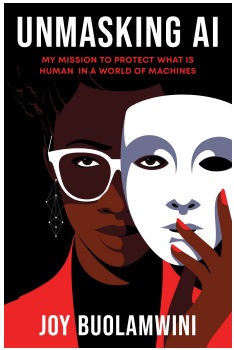
- Infrastructure embeds power relations
  - E.g., a few companies control most of the world's cloud, making smaller countries dependent on them.



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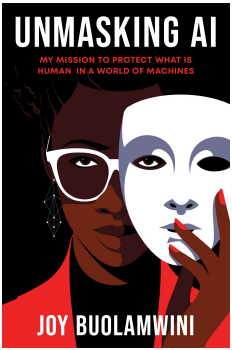
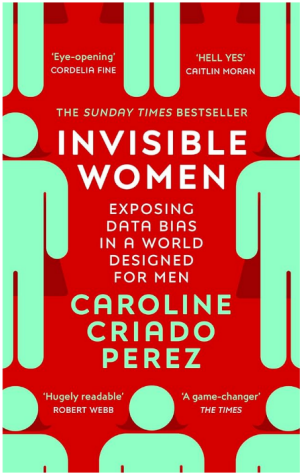
- Infrastructure embeds power relations
  - E.g., a few companies control most of the world's cloud, making smaller countries dependent on them.
- Real costs are more than money
  - E.g., a google data center in the US used over a billion liters of water in one year.





https://tinyurl.com/25fyrz4s

- Healthcare and medical research
  - Women underrepresented in clinical trials, leading to biased treatments.
- Urban planning and public transportation
  - Systems designed for male commuting patterns, neglecting women's needs.
- Workplace design
  - Office temperatures set for men, personal protective equipments often ill-fitting for women.
- Crash test dummies
  - Safety testing primarily based on male bodies, increasing risk for women.
- Smartphone design
  - Devices sized for male hands, making them less user-friendly for women.

Data Feminism

What is Data Feminism? (1/4)

- It is not only about women.
- It is not only for women.
- It is not only about gender.

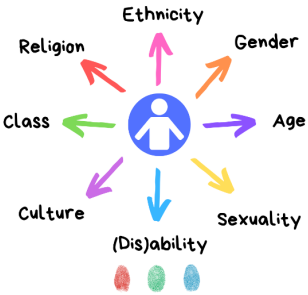
What is Data Feminism? (2/4)

- Data feminism is about power.
- Data is a form of power.
- Power is not distributed equally in the world.

What is Data Feminism? (3/4)

Intersectionality describes:

1. Positionalities as the intersecting aspects of any particular person's identity
2. Intersecting forces of privilege and oppression at work in society.



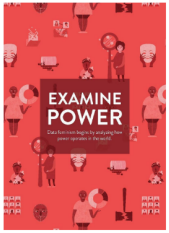
https://tinyurl.com/aedaa2h4

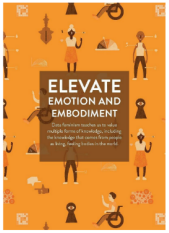
What is Data Feminism? (4/4)

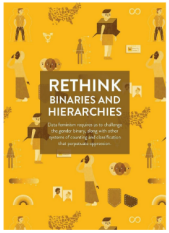
- It is a way of thinking about data, both their uses and their limits, that is informed by direct experience, by a commitment to action, and by intersectional feminist thought.

http://datafeminism.io/blog/book/data-feminism-infographic/

Seven Principles of Data Femiism







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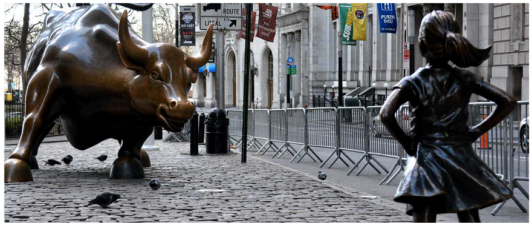
What is Power?

- Structural privilege and structural oppression

Asking Who Questions

- Data science by whom?
- Data science for whom?
- Data science with whose interests and goals?



https://tinyurl.com/mvjnht46

Summary

Summary

- Logic-based AI, Machine Learning, Deep Learning
- Deep Learning models, e.g., DNN, CNN, RNN
- Ciritical view: Data Feminsim

References

- Hands-on machine learning with Scikit-Learn and TensorFlow (Ch. 1, 2)
- Data Feminsim (Introduction)

Questions?